Algorithm 3: Relaxed reachability backpropagation

Input:

- V. Set of nodes
- E_s. Set of scheduled edges (v, u, dep, arr)
- E_t. Set of transfer edges (v, w, Δ)
- country(v). Country associated with node v
- bucket(t). Time bucket mapping function
- $B_{\max}$. Maximum bucket index within 24-hour horizon

Output:

- $R(v, b)$. Set of reachable countries for all $v \in V$, $b \in \{0, \ldots, B_{\max}\}$

// Initialize reachability sets
for all $v \in V$ **do**
    for $b = 0$ to $B_{\max}$ **do**
        $R(v, b) \leftarrow \{\text{country}(v)\}$
// Build relaxed adjacency lists
Initialize $\text{Adj}(v, b) \leftarrow \emptyset$ for all $v \in V$, $b \in \{0, \ldots, B_{\max}\}$
// Scheduled edges
for all $(v, u, dep, arr) \in E_s$ **do**
    $b \leftarrow \text{bucket}(dep)$
    $b' \leftarrow \text{bucket}(arr)$
    if $0 \leq b \leq B_{\max} \wedge 0 \leq b' \leq B_{\max}$ **then**
        $\text{Adj}(v, b) \leftarrow \text{Adj}(v, b) \cup \{(u, b')\}$
// Transfer edges
for all $(v, w, \Delta) \in E_t$ **do**
    for $b = 0$ to $B_{\max}$ **do**
        $t \leftarrow \text{start\_time\_of\_bucket}(b)$
        $t' \leftarrow t + \Delta$
        $b' \leftarrow \text{bucket}(t')$
        if $0 \leq b' \leq B_{\max}$ **then**
            $\text{Adj}(v, b) \leftarrow \text{Adj}(v, b) \cup \{(w, b')\}$
// Backward propagation over time buckets
for $b = B_{\max}$ downto 0 **do**
    repeat
        changed $\leftarrow$ false
        for all $v \in V$ **do**
            for all $(u, b') \in \text{Adj}(v, b)$ **do**
                if $R(u, b') \setminus R(v, b) \neq \emptyset$ **then**
                    $R(v, b) \leftarrow R(v, b) \cup R(u, b')$
                    changed $\leftarrow$ true
    until changed = false
return R